Jacob Lawson

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Education

University of California, Merced — Expected May 2027
B.S. in Computer Science & Engineering — *Merced, CA*

- **Coursework:** Data Structures, Algorithm Design & Analysis, Database Systems, Computer Architecture

Experience

Website & Web App Student Lead — Oct. 2026 – Present
UC Merced – Basic Needs Center — *Merced, CA*

- Own the Basic Needs Center website, shipping UI and content updates to accessibility and mobile standards.
- Partner with staff to scope and ship new features integrating Google Drive, Qualtrics, and Calendly.

Software Engineering Intern (Project Lead) — May 2025 – Jul. 2025
Robotics Cats – Environmental monitoring startup — *California*

- Led a 4-person team building a receipt-digitization pipeline: a React Native (TypeScript) app sends photos to a Flask API that returns store, date, and total as structured JSON for Google Sheets.
- Built the OpenCV preprocessing stage (Hough-line deskew, contour detection, perspective correction, CLAHE) that feeds Gemini 2.5 Flash vision through a schema-constrained prompt.
- Built the progress monitor and docs for LConnect, a Python tool feeding camera snapshots to wildfire detection.

Student Technology Consultant — Aug. 2023 – Oct. 2026
UC Merced – Office of Information Technology — *Merced, CA*

- Resolved 80+ ServiceNow tickets across OS, networking, AV hardware, and SaaS for campus classroom systems.
- Documented recurring failure patterns in Confluence as a first-stop runbook, reducing repeat escalations.

Undergraduate Research Assistant — Oct. 2022 – May 2023
UC Merced – LInC Lab — *Merced, CA*

- Automated cleaning of 600+ CSV files with Python/pandas and standardized data protocols across 4 research projects, cutting processing time 30% and formatting inconsistencies 20%.

Projects

OWCS Muse | *Python, FastAPI, PostgreSQL, React, Docker, GitHub Actions, Fly.io, Vercel* — Live

- Built and launched a plain-English stats search engine for Overwatch esports: a FastAPI service parses each question into a validated structured query, then compiles it to parameterized SQL over a Postgres warehouse.
- Engineered a resumable, idempotent nightly ingest reconciling the FACEIT and Liquipedia APIs, archiving every raw payload for offline reparsing and resolving team and player identity across sources.
- Guarded correctness with 600+ pytest tests, a 94-case golden suite of hand-verified answers, and 18 production invariant checks; CI runs against Postgres 16 and 17.

Bobcat Bulletin | *Python, FastAPI, SQLAlchemy, React, Docker, GitHub Actions, Fly.io* — Live | GitHub

- Turned UC Merced PD's daily activity logs into a searchable, filterable dashboard of 23,000+ incidents.
- Moved scraping to a twice-daily GitHub Actions job posting to the REST API, keeping the backend VM at 256 MB.

LiftLabs | *React, Vite, Flask, SQLAlchemy, PostgreSQL, Vercel, Render* — Live

- Built a fitness forum with JWT auth, role-based moderation, and server-validated one-vote-per-user enforcement.
- Deployed the SPA and API across Vercel and Render, resolving cross-origin auth and persistent file storage issues.

Greenview | *JavaScript, Chrome Extension API, Manifest V3* — GitHub

- Shipped a Chrome/Firefox extension tagging job links by ATS, using MutationObserver for async-loaded listings.

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, C++, Go, PowerShell, HTML/CSS
Frameworks: FastAPI, Flask, SQLAlchemy, React, React Native, Node.js, Next.js, Tailwind, pandas, OpenCV, pytest
Infrastructure: PostgreSQL, Supabase, Docker, GitHub Actions (CI/CD), Fly.io, Vercel, Render, AWS, Linux, Git
Concepts: REST API design, data pipelines/ETL, relational data modeling, automated testing, authentication